

Name: _____

Problem Set 1

(Questions 1-4 for submission; the rest will be discussed in TA session)

(1) Consider the following prices (in Shekels) for a phone sold in Israel. The retail price is 3000. Prior to selling the phone to the consumer, the retailer pays 2500 to an importer. The importer's costs are 1700 that are paid to the manufacturer in China. The Chinese manufacturer uses intermediate goods that were purchased in Vietnam for 500 and other intermediate goods that were purchased in China for 300.

- (a) What is the Israeli retailer's value added?**
- (b) What is the Israeli importer's value added?**
- (c) What is the Chinese manufacturer's value added?**
- (d) How much of the tablet's retail value show up in Israel's GDP according to the product (value added approach)? How much in China's GDP?**
- (e) Where does the tablet show up in the expenditure components of Israel's GDP? How much of the tablet's retail value show up in Israel's GDP according to the expenditure approach?**
- (f) If we count Israel's net exports (or trade balance) with each individual country, how does the tablet affect Israel's trade balance with China? With Vietnam?**

2) Consider an economy that produces three products: Apples, Oranges, and machines. Production and price data are:

	Year	Quantity	Price
Apples	2010	8,000kg	\$2 per kg
	2020	10,000kg	\$4 per kg
Oranges	2010	3,000kg	\$6 per kg
	2020	6,000kg	\$3 per kg
Machines	2010	100	\$300 each
	2020	110	\$400 each

- (a) What is the growth rate of nominal GDP from 2010 to 2020?**
- (b) What is the growth rate of real GDP at 2010 prices?**
- (c) What is the growth rate of real GDP at 2020 prices?**
- (d) What is the GDP deflator in 2010 and 2020 if you calculate real GDP based on 2010 prices?**
- (e) What is the GDP deflator in 2010 and 2020 if you calculate real GDP based on 2020 prices?**
- (f) What is the inflation rate (based on GDP deflator) associated with the price levels of (d), (e)**

(3) Continue with the same data as in question #2. Now let's calculate Consumer Price Index inflation in the economy. Assume that there are 1,000 households identical households in the economy, and each of them consumes the same basket of consumption goods (i.e only the fruit, not the machines, enter the CPI calculations).

(a) Assume that the "representative consumption basket" is the one reflected by the quantities of 2010. What is the price of this basket in 2010? 2020?

(b) Assume that the "representative consumption basket" is the one reflected by the quantities of 2020. What is the price of this basket in 2010? 2020?

(c) What is the inflation rate associated with the prices in (a), (b) ?

(d) Why are the inflation rates different?

(e) What is the reason for the different inflation rates in this question relative to the inflation rates based on GDP deflator?

(4) Calculating average growth rates. The difference between GDP and GDP per capita

The Penn World Table (PWT) is a database with information on various economic variables for more than 180 countries and over time. The strength of this database is that it is a serious attempt to have comparable figures across countries and over time.

You can download the recent version (11.0) here:

<https://www.rug.nl/ggdc/productivity/pwt/?lang=en>

Pick any two countries that you like. For each country use (i) the rgdpna series (Real GDP at constant 2017 national prices (in mil. 2017US\$)); and (ii) the pop (population in millions) series.

Divide real GDP by population, so that you have a series of real GDP per capita for each country.

- (i) What is the average growth rate of GDP for each of the two countries between 1990 and the latest year that you have (for most countries this would be 2022 or 2023, but for some the most updated may be an earlier year)?
- (ii) What is the average growth rate of GDP *per capita* for each of the two countries between 1990 and the latest year that you have?
Which growth rate is larger: GDP or GDP per capita? In case of a difference, why do you think it exists?
If you were asked to describe (to a friend or a relative) the growth experience of a country, would you use GDP or GDP per capita? Briefly explain.
- (iii) Now calculate the average growth rates of GDP per capita by decade for each country (see table below), and briefly describe how different these growth rates are relative to your calculations in part (ii)

Period	Country 1	Country 2
1990-1999		
2000-2009		
2010-2019		

(5) (For TA session; all number in the following example are invented)

Assume that in 2015 China's GDP per capita was 8,000 yuan, and that the market exchange rate between the yuan and the U.S. dollar was traded for 8 yuans per dollar.

Assume that in the same year Albania's GDP per capita was 20,000 Albanian Leks (that's the name of the currency) and the U.S. dollar was traded for 4 Leks per dollar.

Also assume that a representative product costs \$1 in the US, 4 yuans in China, and 10 Leks in Albania.

(a) What is the market value of China's GDP in U.S. dollars? What is the market value of Albania's GDP in US dollars?

(b) What is the "purchasing power exchange rate" between the yuan and U.S. dollar? What is the "purchasing power exchange rate" between the Albanian Lek and U.S. dollar?

(c) What is the China's PPP GDP per capita? What is Albania's PPP GDP per capita?

(d) Which country looks richer according to the calculation in part (a)? and according to the calculation in part (c)? Explain the source of the difference and why it is usually better to use the PPP measure rather than the nominal one.

(e) Give one example for a measurement problem when calculating the PPP exchange rate.

(6) For TA session: Growth rates in the Data and how they are reported.

Go to <https://fred.stlouisfed.org> and search for real GDP, so that you can see the quarterly real gross domestic product series for the US. Use the data points for the second quarter of 2020 and the third quarter of 2020 to calculate the growth rate.

When the data on GDP growth for the third quarter of 2020 was released, the “headline number” reported was around 33% GDP growth. Is this identical to your calculation? Explain.

(7) (For TA session: Related to questions 2 and 3, but with real data). The federal reserve bank of St. Louis maintains a data source called FRED, with a lot macro data for the US and other countries. You can check it out here:

<https://fred.stlouisfed.org>

Using FRED, download two data series: (i) Consumer Price Index: Total All Items for the United States; and (ii) Gross Domestic Product: Implicit Price Deflator.

For both series, use quarterly and seasonally adjusted data. (Use data starting from the earliest date possible).

a. download the data and calculate the inflation rate based on CPI and the inflation rate based on the GDP deflator.

b. plot the two inflation rates (i.e. put time period on the x-axis and the price index on the y-axis). Are there any noticeable differences?

c. Compute the correlation coefficient between the two series of inflation. How similar are they? Explain why you expect them to be similar or different.

(note that in order to calculate the correlation you will have to consider the same time period for both data series. You can also search for a brief description of the correlation coefficient).

d. Explain why it is important to use seasonally adjusted data when calculating inflation rates. (To illustrate, you can search for the series Consumer Price Index for all Urban Consumers: all items. You will see that this series is available with and without seasonal adjustment. If you zoom in on a relatively short period of a few years you will be able to see the differences)

(8) (For discussion in TA session) In class we described the reasoning behind the equation for the expected real interest rate

$$E[r] = i - \pi^e$$

Where $E[r]$ is the expected real interest rate, i is the nominal interest rate, and π^e is expected inflation.

Sometimes it is useful to use data on market prices to calculate the expected inflation according to financial markets. In principle, this can be done by rearranging the equation

$$\pi^e = i - E[r]$$

If we have market prices for the nominal and expected real rate, the calculation is easy. One such price involved indexed vs. non-indexed government bonds. In the US, the treasury issues “regular” bonds that are not indexed, as well as “Treasury Inflation-Protected Securities” (TIPS).

- (a) Explain why a regular bond can be used as a proxy for the nominal rate, and the indexed bond can be used as a proxy for real rates.
- (b) Explain why it is important that the two bonds will be issued by the same borrower and have similar maturities.
- (c) Go to <https://fred.stlouisfed.org> (a very useful resource for gathering and visualizing macroeconomic data). Search for 5-year, 10-year, and 30-year Breakeven Inflation Rate (three separate series). These are “...what market participants expect inflation to be in the next 5/10/30 years, on average.” Plot the three series from the earliest available point. (You can download figures from FRED or download data and generate the plots on your own).
- (d) Do the three appear to be positively correlated?
- (e) Focusing on the recent spike around 2022: why do you think that the 5-years expectations are higher than the other two? Why do you think that the expected inflation is lower than what inflation was? Why do you think that over the longer-term inflation is expected to be close to 2%?